SEEING SOUNDS, MISSING MEANINGS: A CHINESE–ENGLISH BILINGUAL BENCHMARK FOR VISUAL HOMOPHONE REASONING

Peijin Xie[1], Jiayi Zhao[2], Xiaoxue Wang[2], Zhen Xu[2], Baoxun Wang[2], Bingquan Liu[1]

[1]Harbin Institute of Technology, ITNLP Lab, Harbin, China
[2]Tencent, Platform and Content Group, QQAI, Beijin, China

ABSTRACT

Visual homophone rebuses encode expressions through images whose concepts evoke similar-sounding words. Solving them requires visual grounding, lexical selection, phonological association, and answer composition. We introduce **BiRebus**, a Chinese–English benchmark containing approximately 1,600 emoji and pictorial rebuses. To construct diverse puzzles at scale, we develop an agentic pipeline that integrates social-media collection, homophone decomposition, visual planning, image generation, iterative verification, and human review. On a challenging diagnostic split, the strongest evaluated model, GPT-6-Astra, achieves only 58% strict accuracy under direct prompting, compared with 63% for human solvers. We further propose a structured inference strategy based on *visual grounding*, *phonological imagination*, and answer verification. Without parameter updates, it raises aggregate model accuracy from 48% to 56%. BiRebus and its agentic construction pipeline will be released at `https://github.com/xiepeijinhit-hue/rebus`.

***Index Terms*—** Vision-Language Models, Visual Homophone Reasoning, Rebus Puzzles, Multimodal Reasoning, Bilingual Benchmark

1. INTRODUCTION

Vision-language models (VLMs) perform strongly on visual question answering and multimodal reasoning. Early benchmarks emphasize grounding and compositional reasoning [1, 2, 3], while recent work covers broader perceptual, knowledge-intensive, and expert-level capabilities [4, 5, 6, 7, 8, 9, 10]. However, whether VLMs can connect visual concepts to implicit linguistic meanings through pronunciation remains underexplored.

Visual homophone rebuses provide a compact test of this capability. They represent an expression using visual concepts whose names resemble parts of the intended answer in sound. Solving them requires a model to identify critical clues, choose suitable names, recover phonological associations, and compose a valid expression. The task therefore tests not only what a model sees, but also how it names and pronounces what it sees.



Fig. 1. Representative Chinese and English rebuses in **BiRebus**. Emoji provide constrained clues, whereas open-ended scenes require answer-relevant visual abstraction and lexical selection.

Figure 1 illustrates the two visual formats in BiRebus. A pair of dice evokes "pair of dice," which sounds like "paradise," while a Chinese emoji sequence encodes *zhang deng jie cai* through near-homophonic concepts. Emoji provide relatively constrained meanings, whereas pictorial scenes admit many plausible descriptions, only a few of which activate the intended sounds.

Unlike conventional visual question answering, BiRebus provides no explicit inventory of relevant objects or candidate answers. A solver must infer both the visual units and the linguistic granularity at which they should be interpreted. The benchmark therefore tests whether multimodal representations remain reliable when perception must support an indirect, pronunciation-mediated objective.

Existing puzzle benchmarks examine reasoning beyond literal recognition. Iconary studies drawing-based communication [11], and BRAINTEASER evaluates lateral reasoning in textual puzzles [12]. REBUS and COLUMBUS investigate visual wordplay, while Emoji2Idiom studies cryptic emoji interpretation [13]. However, existing resources primarily focus on English, constrained symbols, or template-based layouts. They do not systematically compare bilingual homophone reasoning across standardized symbols and open-ended scenes.

This comparison is important because homophonic wordplay is language dependent. Chinese rebuses often use

character-level homophones, near-homophones, four-character idioms, and culturally specific expressions. English rebuses more often rely on object-name homophones, syllable segmentation, and phonetic blending. Such puzzles cannot be transferred reliably through direct translation.

We introduce **BiRebus**, a unified Chinese–English benchmark for visual homophone reasoning. It contains approximately 1,600 emoji and pictorial rebuses collected from public content or produced by an agentic synthesis pipeline. To the best of our knowledge, BiRebus is the first benchmark to evaluate Chinese and English visual homophone reasoning under a unified protocol.

On a balanced diagnostic split of 400 high quality examples, the strongest VLM achieves only 58% strict accuracy under direct prompting. Models frequently recognize visible elements but choose unsuitable names or fail to activate the intended sound associations. A structured strategy based on visual grounding, phonological imagination, and answer verification improves aggregate accuracy from 48% to 56% without parameter updates.

Our contributions are threefold:

- We introduce **BiRebus**, a Chinese–English benchmark of approximately 1,600 emoji and pictorial rebuses for evaluating visual homophone reasoning.
- We develop an **agentic construction pipeline** combining public-content collection, language-specific homophone decomposition, image generation, iterative verification, and human review.
- We identify the visual-to-phonological transition as a major bottleneck and propose a structured strategy that improves aggregate strict accuracy by 8 percentage points.

2. AGENTIC DATA GENERATION

BiRebus is constructed through two complementary routes: collecting human-created rebuses from public social-media content and synthesizing new puzzles through an agentic pipeline. As shown in Fig. 2, both routes share a final quality-control stage for visual clarity, solvability, and image–answer consistency.

2.1. Social-Media Collection

A collection agent retrieves public Chinese and English rebuses using language-specific keywords and engagement signals. It extracts puzzles and answers, removes duplicates and near-duplicates, and filters advertisements, answer leakage, and samples with insufficient visual evidence. Each puzzle is retained in its original language because its wordplay may not survive translation. Human annotators verify the language, answer, critical clues, and phonological mechanism.

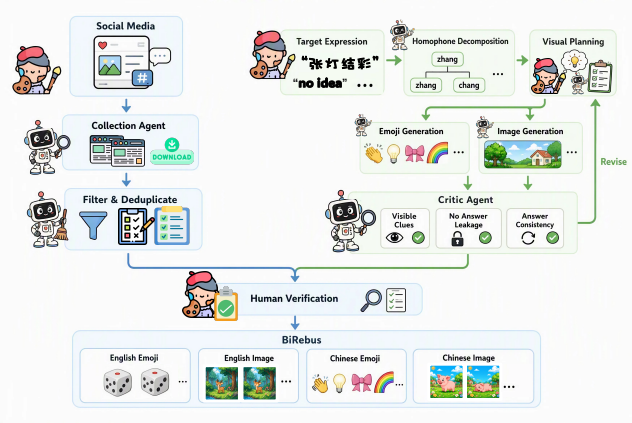


Fig. 2. Construction pipeline for BiRebus. A collection agent retrieves public rebuses, while a synthesis branch converts target expressions into emoji or pictorial puzzles. Agentic verification and human review ensure clue visibility, solvability, and image–answer consistency.

2.2. Agentic Rebus Synthesis

To increase visual diversity, we develop a text-to-rebus pipeline for both visual formats. Given a target word, phrase, or idiom, a planning agent decomposes it into homophonic or near-homophonic units and maps each unit to a depictable object, attribute, action, or relation. For emoji rebuses, it selects a minimal sequence of standardized symbols. For pictorial rebuses, it converts the clues into a scene plan and invokes GPT-Image-2 to generate a cartoon-style or pixel-art image.

To improve dataset diversity, synthesis targets span common nouns, phrases, idioms, and culturally grounded expressions. We also vary clue count, spatial arrangement, and phonetic similarity. These controls reduce dependence on a single visual template and enable future analysis of how puzzle complexity interacts with language and visual format.

Generation follows a generate–criticize–revise loop related to self-feedback methods [14, 15]. A critic checks whether all intended clues are visible, whether unintended objects or text leak the answer, and whether the depicted concepts support the intended sounds. Failed candidates are returned for replanning or regeneration.

Human annotators review low-confidence cases and agent disagreements. They may correct annotations, request regeneration, or reject a sample. This agent–human workflow enables scalable construction while preserving visual and linguistic quality.

Table 1. Strict accuracy (%) on the diagnostic split. VLM results use direct prompting.

Model	En-E	En-P	Zh-E	Zh-P	All
Human	**67**	**52**	**85**	48	**63**
GPT-6-Astra	43	46	74	**68**	58
Gemini-3.8-F	32	42	57	59	48
GPT-5.6-Sol	30	34	67	49	45
DeepSeek-V4-V	27	27	69	49	43

3. EXPERIMENTS AND ANALYSIS

3.1. Evaluation Setup

We evaluate four representative VLMs on four subsets: **English Emoji** (En-E), **English Pictorial** (En-P), **Chinese Emoji-Idiom** (Zh-E), and **Chinese Pictorial** (Zh-P). From approximately 1,600 examples, we select a balanced diagnostic split of 400, with 100 per subset, for two reasons.

First, this selection improves evaluation quality. Annotators verify visual–concept consistency and whether the clue pronunciations compose into the reference answer. Among valid candidates, they prioritize examples with a stronger contrast between the literal visual semantics and the intended expression, making them more diagnostic of pronunciation-mediated reasoning. Second, using 100 examples per subset controls for subset size and enables direct comparison across languages and visual formats. The split is therefore designed for controlled diagnostic analysis rather than for estimating performance on the full benchmark.

Each model predicts the encoded expression in its original language. A prediction is correct only if it exactly matches the reference after normalizing capitalization, whitespace, hyphens, and surrounding punctuation. Accuracy is computed over all 100 examples per subset and all 400 examples overall. We use temperature 0 and a maximum output budget of 1,600 tokens.

3.2. Direct-Prompting Results

Table 1 reports direct-prompting results and the performance of independent human solvers. GPT-6-Astra is the strongest VLM at 58%, followed by Gemini-3.8-Flash, GPT-5.6-Sol, and DeepSeek-V4-Flash-Vision at 48%, 45%, and 43%. Humans achieve 63%, exceeding the strongest VLM by 5 points while confirming the difficulty of the split.

The VLMs perform better on the Chinese subsets, although differences in sources, answer spaces, and wordplay mechanisms prevent attributing this pattern solely to multilingual ability. Human accuracy also varies from 48% on Zh-P to 85% on Zh-E. GPT-6-Astra exceeds humans on Zh-P but remains behind on the other three subsets. The low English scores further indicate that object recognition alone is insufficient: models must select clue names whose sounds form a valid target expression.



Fig. 3. Structured reasoning for "eye + scream" → "ice cream." The model grounds visual concepts, explores sound associations, and verifies the final answer against all clues.

The results also show that recognition accuracy does not necessarily translate into stronger rebus reasoning. A model may identify every depicted entity yet fail because its preferred lexical labels do not match the required sounds. Conversely, an imperfect description may produce the correct answer if it preserves phonologically useful clues. This mismatch motivates stage-aware analysis alongside end-to-end accuracy.

3.3. Structured Prompting and Results

Direct prompts often elicit literal descriptions rather than sound-based answers. Motivated by intermediate reasoning methods [16, 17], we introduce the three-stage strategy in Fig. 3. **Visual grounding** (VG) extracts essential objects, attributes, counts, actions, and relations. **Phonological imagination** (PI) explores alternative clue names, homophones, and near-homophones. The model then composes candidate expressions and verifies their coverage of the visual clues.

The English prompt emphasizes sound blending and word-level composition, whereas the Chinese prompt emphasizes character-level homophonic or near-homophonic substitution. Models may generate intermediate reasoning but must place the final prediction after a fixed `ANSWER:` marker.

As shown in Table 2, VG+PI improves every subset on average and raises aggregate accuracy from 48% to 56%. En-P obtains the largest gain at 11 points.

GPT-5.6-Sol gains 16 points overall, followed by GPT-6-Astra and DeepSeek-V4-Flash-Vision with gains of 12 and 5 points. These improvements suggest that many direct-prompting errors arise from failing to activate the required reasoning procedure rather than from a complete absence of visual or lexical knowledge.

Gemini-3.8-Flash decreases by 1 point. Its longer traces occasionally introduce formatting errors or fail to converge on one answer. Structured prompting is therefore broadly ef-

Table 2. Strict accuracy (%) with Direct and VG+PI prompting. Each entry is Direct/VG+PI.

Model	En-E	En-P	Zh-E	Zh-P	All
GPT-6-Astra	43/58	46/63	74/81	68/78	58/70
Gemini-3.8-F	32/27	42/43	57/59	59/57	48/47
GPT-5.6-Sol	30/49	34/51	67/79	49/65	45/61
DeepSeek-V4-V	27/30	27/35	69/75	49/52	43/48
Aggregate	33/41	37/48	67/74	56/63	48/56
Gain	**+8**	**+11**	**+7**	**+7**	**+8**

Table 3. Overall strict accuracy (%) in the ablation study.

Model	Direct	VG	VG+PI
GPT-6-Astra	58	65	70
GPT-5.6-Sol	45	55	61
Aggregate	51	60	66
Increment	–	+9	+6

fective, but its benefit depends on instruction following and output control.

3.4. Ablation and Bottleneck Analysis

We evaluate GPT-6-Astra and GPT-5.6-Sol with VG alone to separate the contributions of grounding and phonological search. Table 3 shows that VG raises their accuracy from 58% to 65% and from 45% to 55%. Adding PI further increases performance to 70% and 61%. Across both models, VG contributes 9 points and PI another 6 points.

The larger VG gain identifies visual abstraction and clue selection as an initial bottleneck. Models may recognize an object but choose a name that is too broad, specific, or phonologically unhelpful. PI addresses a downstream problem: a grounded clue may admit several lexicalizations or pronunciations, only one of which supports the target expression.

To examine the remaining errors, we assign each incorrect VG+PI prediction to its earliest failed stage. A **VG failure** omits or misidentifies a critical clue; a **PI failure** grounds the clues but misses the intended homophone; and a **composition/verification failure** recovers relevant clues and sounds but combines them incorrectly or leaves part of the image unexplained. The criterion avoids counting cascading errors.

Figure 4 reveals format-dependent bottlenecks. PI accounts for 54% of En-E errors and 48% of Zh-E errors, whereas only 19% and 18% on VG. Emoji constrain visual interpretation but retain lexical and phonological ambiguity.

Grounding is prominent for pictorial rebuses, accounting

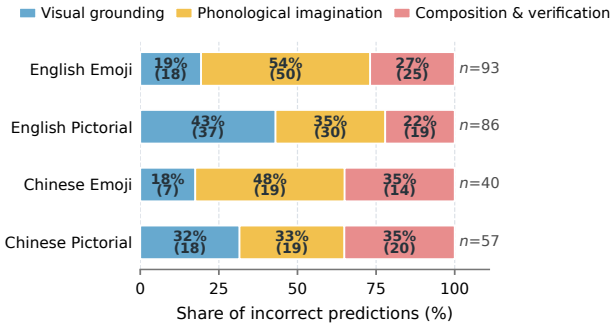


Fig. 4. Failure-stage distributions among incorrect VG+PI predictions. Results are aggregated across two models and normalized within each subset's errors.

for 43% of En-P errors and 32% of Zh-P errors. Open-ended scenes admit more descriptions and distractors, increasing risk of selecting phonologically unhelpful concepts. Composition and verification account for approximately 35% of errors on both Chinese subsets, compared with 27% on En-E and 22% on En-P. This may reflect the global constraints of Chinese idioms, where plausible character-level homophones must form a valid expression in the correct order.

The experiments provide complementary evidence. The table measures gains from supporting each stage, whereas the figure locates errors remain after the full strategy. Improving an early stage may expose downstream failures, supporting a cascading account of visual homophone reasoning.

Finally, replacing model-generated descriptions with human-provided concepts yields 59% accuracy on En-P but limited and inconsistent effects on En-E. This supports a stronger grounding bottleneck in open-ended scenes while showing that human concepts alone cannot resolve lexical and phonological ambiguity. Because this oracle experiment covers only the English subsets and two models, we treat it as a diagnostic intervention rather than an upper bound.

4. CONCLUSION

We introduced **BiRebus**, a Chinese–English benchmark containing approximately 1,600 emoji and pictorial rebuses for evaluating visual homophone reasoning. The benchmark combines public-content collection with an agentic generation and verification pipeline.

On a challenging diagnostic split, the strongest VLM achieves 58% strict accuracy under direct prompting, compared with 63% for humans. Models often recognize visible content but fail during lexical selection or phonological association. A structured strategy based on visual grounding, phonological imagination, and answer verification raises aggregate model accuracy from 48% to 56% without parameter updates. BiRebus thus provides a challenging testbed for phonologically grounded multimodal reasoning.

5. REFERENCES

[1] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh, "Making the v in vqa matter: Elevating the role of image understanding in visual question answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 6904–6913.

[2] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick, "CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 2901–2910.

[3] Drew A. Hudson and Christopher D. Manning, "GQA: A new dataset for real-world visual reasoning and compositional question answering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 6700–6709.

[4] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, Rongrong Ji, Caifeng Shan, and Ran He, "Mme: A comprehensive evaluation benchmark for multimodal large language models," 2025.

[5] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, Kai Chen, and Dahua Lin, "MMBench: Is your multi-modal model an all-around player?," in *European Conference on Computer Vision*, 2024, pp. 216–233.

[6] Bohao Li, Rui Wang, Guangzhi Wang, Yuying Ge, Yixiao Ge, and Ying Shan, "SEED-Bench: Benchmarking multimodal large language models with generative comprehension," *arXiv preprint arXiv:2307.16125*, 2023.

[7] Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang, "MM-Vet: Evaluating large multimodal models for integrated capabilities," *arXiv preprint arXiv:2308.02490*, 2023.

[8] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, et al., "MMMU: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9556–9567.

[9] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao, "MathVista: Evaluating mathematical reasoning of foundation models in visual contexts," in *International Conference on Learning Representations*, 2024.

[10] Pan Lu, Swaroop Mishra, Tony Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan, "Learn to explain: Multimodal reasoning via thought chains for science question answering," in *Advances in Neural Information Processing Systems*, 2022, vol. 35, pp. 2507–2521.

[11] Peter Clark, Oyvind Tafjord, and Kyle Richardson, "Iconary: A pictionary-based game for testing multimodal communication with drawings and text," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 2021, pp. 1864–1874, Association for Computational Linguistics.

[12] Yifan Jiang, Filip Ilievski, Kaixin Ma, and Zhivar Sourati, "BRAINTEASER: Lateral thinking puzzles for large language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 2023, Association for Computational Linguistics.

[13] Koen Kraaijveld, Yifan Jiang, Kaixin Ma, and Filip Ilievski, "Columbus: Evaluating cognitive lateral understanding through multiple-choice rebuses," 2024.

[14] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Sean Welleck, Bodhisattwa Prasad Majumder, Shashank Gupta, Amir Yazdanbakhsh, and Peter Clark, "Self-refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems*, 2023, vol. 36.

[15] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems*, 2023, vol. 36.

[16] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems*, 2022, vol. 35, pp. 24824–24837.

[17] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou, "Self-consistency improves chain of thought reasoning in language models," in *International Conference on Learning Representations*, 2023.